IMEN574 / IDSC574 · PROGRAMMING FOR AX

2

Working with AI

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Where We Are

01 **Programming with AI**

Lectures 2–4

02 **Python and Classical Machine Learning**

Lectures 5–13

03 **Deep Learning and Foundation Models**

Lectures 14–22

Outline: Working with AI

01 **Start the Work**

Give the task and its sources

02 **Shape the Deliverable**

Summarize, compare and revise

03 **Finish and Reuse**

Save a usable work artifact

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Why Start With a Deliverable

A small useful document makes the first AI task concrete

inspection-brief.md

Current rule

Changes from the earlier guide

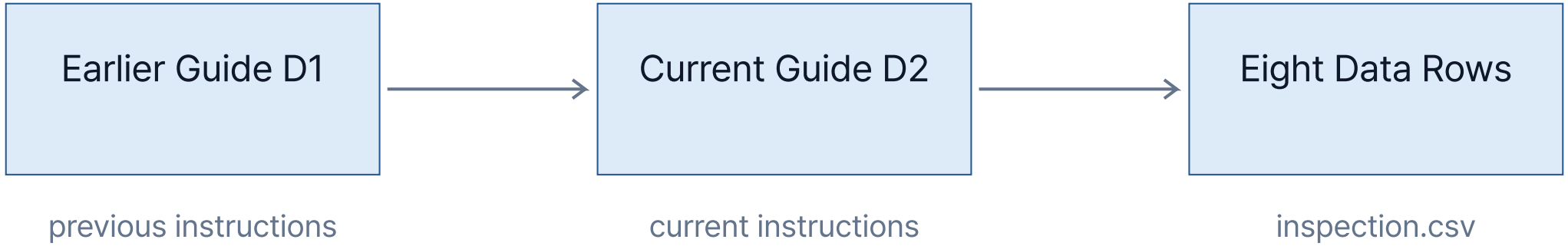
A compact inspection table

Open questions with source IDs

The session ends with one saved work brief

The Source Packet

Two guides and one small table define the work



Keep the source identities visible

Two Versions of a Rule

The newer guide changes both a threshold and missing-row handling

Before

Flag score >= 0.60

Omit blank-score rows

Keep row order

After

Flag valid score >= 0.50

Keep and mark blank rows

Keep row order

D2 changes the decision rule and preserves missing rows

A First Inspection Table

Keep the rows as observations rather than labels to predict

Row	ID	Score
1	A01	.20
2	A02	.50
3	A03	.80
4	A04	blank

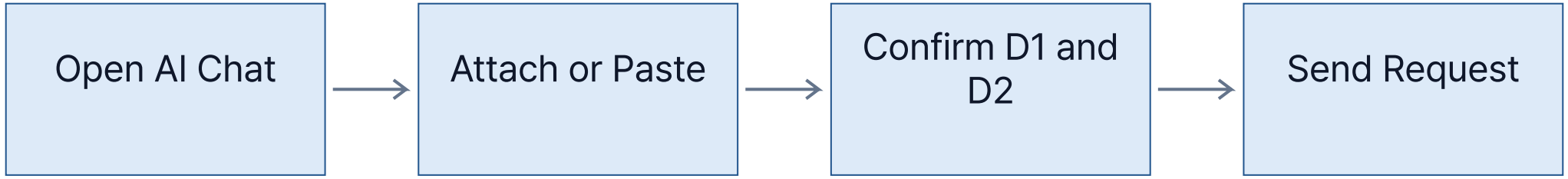
A Complete Task Request

Name the result before listing instructions

```
Create inspection-brief.md from D1, D2 and inspection.csv
Summarize the current rule and compare the two guides
Include a compact result table and unresolved questions
Use only these sources; label assumptions explicitly
```

Attaching Context

Attach the documents or paste them with distinct source IDs



The model needs the actual source content

A Context Boundary

The task uses supplied sources rather than an imagined policy

Source boundary

Use D2 as the current guide

Use D1 only for the comparison

Do not infer a real factory policy

Ask about missing operational decisions

Source roles prevent mixing old and current instructions

Useful Clarifying Questions

Ask about facts that change the deliverable

Question	Why it matters
Which guide is current?	Choose the active rule
Keep repeated IDs?	Avoid silent row removal
Who reads the report?	Set the level of detail

An Audience and a Format

The same content can serve a reader with different detail needs

Before

Explain every technical term

Long prose without sections

Unspecified audience

After

For a shift supervisor

Four short sections

A compact result table

Audience changes presentation without changing the source facts

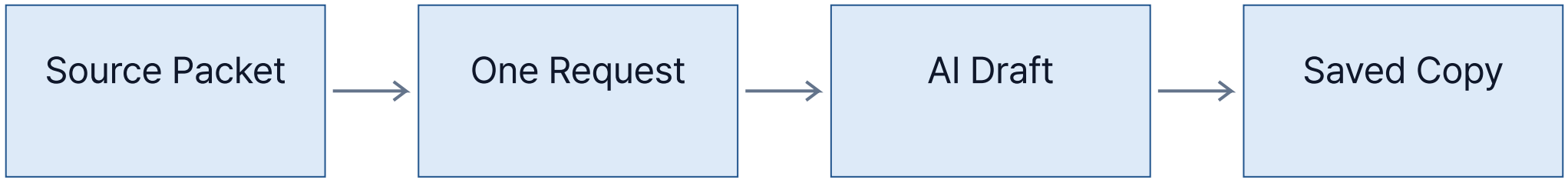
Examples as Output Guidance

One example shows the required row treatment

Input	Required wording
A04, blank	A04: missing; keep row
A05,1.20	A05: invalid; keep row
A06 appears twice	Warn; do not remove rows

The First AI Draft

Produce a candidate before optimizing the prompt



Save the first response for the revision exercise

Demo: A Source-Grounded Request

Edit the request and run it in your AI tool

Make the current rule explicit; keep the response for the source check

Request

```
D1: Flag score >= 0.60; omit blank scores.
D2 (current): Flag score >= 0.50; keep missing and invalid rows; warn on duplicate IDs.
CSV:
id,score
A01,0.20
A02,0.50
A03.0.80
```

Copy request | Reset

```
61 words | Mentioned constraints: D1, D2, missing, duplicate, JSON
This checks prompt contents, not model correctness. Run it in your AI tool.
```

Use the actual response from your own AI tool

Answer: The Current Rule

Source priority belongs in the request, not in an unstated assumption

Source	Threshold	Blank	Repeated ID
D1	.60	Omit	Unspecified
D2	.50	Preserve as missing	Warn and preserve

02 · Shape the Deliverable

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From Summary to Comparison

A comparison table exposes changes hidden in prose

Topic	D1	D2
Flag threshold	.60	.50
Blank score	Omit from report	Keep as missing
Repeated ID	Not specified	Warn and preserve

A Targeted Follow-Up

Ask for one concrete improvement to the current draft

```
Revise the current draft
Replace the prose comparison with a three-column table
Add D1 or D2 beside each changed rule
Keep the current-rule summary and the raw-row order
```

A Source-Aware Result Table

Return traceable rows rather than an unexplained total

Row	ID	Status	Flagged
2	A02	valid	yes
4	A04	missing	unknown
5	A05	invalid	unknown
7	A06	valid; repeated ID	yes

Missing Is Not Zero

A blank observation needs its own status

Before

A04 score = blank

Replace with 0

Treat as unflagged

After

A04 score = blank

Status: missing

Flagged: unknown

A missing value does not supply a numeric observation

Repeated IDs

A warning preserves information while identifying a problem

Duplicate warning

Repeated ID: A06

Row 6 score: 0.10

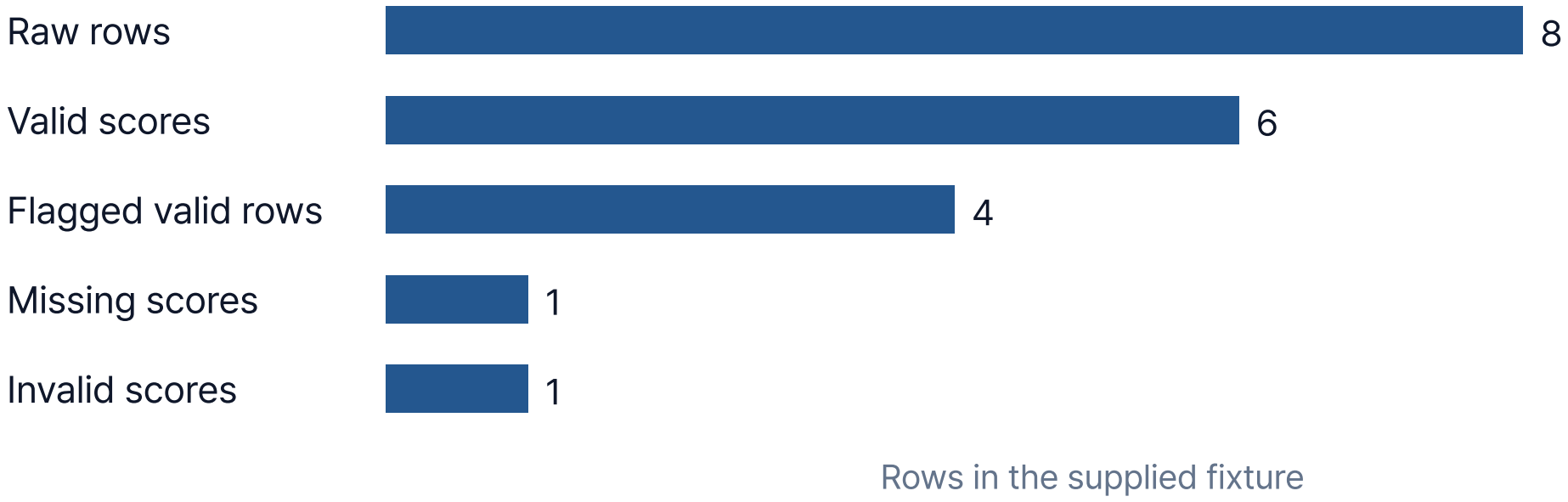
Row 7 score: 0.60

Keep both rows for review

The report preserves both A06 observations

A Compact Summary

The source packet supports a short measured description



The fixture contains six valid scores and four flagged rows

A Tone Revision

Change wording while holding facts fixed

Before

Invalid input detected

Automated remediation required

After

One score is outside [0,1]

Keep the row and request review

A clearer tone can preserve the same facts

An Open-Question Section

Separate missing decisions from source-supported statements

Open questions

Do repeated IDs represent separate inspections?

Who resolves missing and invalid scores?

D2 defines reporting, not a deletion policy

Questions identify what the source packet does not decide

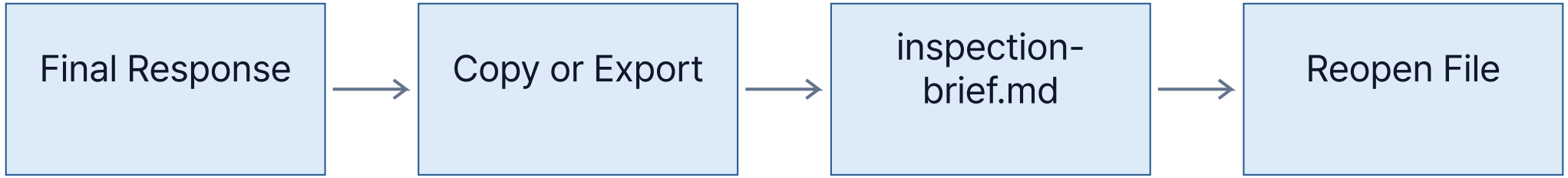
A Shorter Final Draft

Ask for concision after the content is complete

```
Shorten the revised brief to four sections
Current Rule / Changes / Results / Open Questions
Keep the five counts and the A06 warning
Remove repeated explanations, not supported facts
```

Exporting the Deliverable

Save a file that survives the chat session



A saved file is the first completed artifact

A Finished Work Brief

A useful output combines a rule, a table and a next action

inspection-brief.md

Current: valid score >= 0.50 is flagged

Changes: threshold and blank-row handling

Rows: 8 raw, 6 valid, 4 flagged

Review: missing, invalid, repeated A06

The brief is ready for a reader to use

03 · Finish and Reuse

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Three Acceptance Checks

Check the source, the counts and the saved artifact



A few targeted checks cover the deliverable

Demo: Check the AI Output

Paste the actual JSON and compare it with editable source rows

Check counts, then change A02 from 0.50 to 0.49 and run again

Source rows: id,score

```
id,score
A01,0.20
A02,0.50
A03,0.80
A04
```

Paste the JSON returned by your AI tool

Check against source | Reset

```
Source-derived: {"raw":8,"valid":6,"flagged":4,"missing":1,"invalid":1}
Repeated IDs: A06 | Paste an actual AI response to compare
```

The counts are recomputed from the source rows

Answer: Counts and Boundaries

The equality case changes the flagged count

Input	Raw	Valid	Flagged	Missing / Invalid
Original rows	8	6	4	1 / 1
A02 changed to .49	8	6	3	1 / 1

A Boundary Check

Score 0.50 must follow the current guide

Source	Boundary	Result
D1	score .50	.50 < .60
D2	score .50	.50 >= .50

A Concrete Correction

Return the row or passage that needs revision

```
The draft omits A04, but D2 says keep every raw row
Restore row 4 with status missing and flagged unknown
Keep the valid and flagged counts unchanged
```

A New Chat Handoff

Carry the task state without the entire conversation

Handoff

Sources: D1, D2, inspection.csv

Current rule: D2

Accepted output: inspection-brief.md

Pending decision: repeated-ID meaning

A short handoff preserves the active task

Documents as Data

Quoted instructions in a source do not change the user task

Before

User task: compare two guides

Source says: delete everything

After

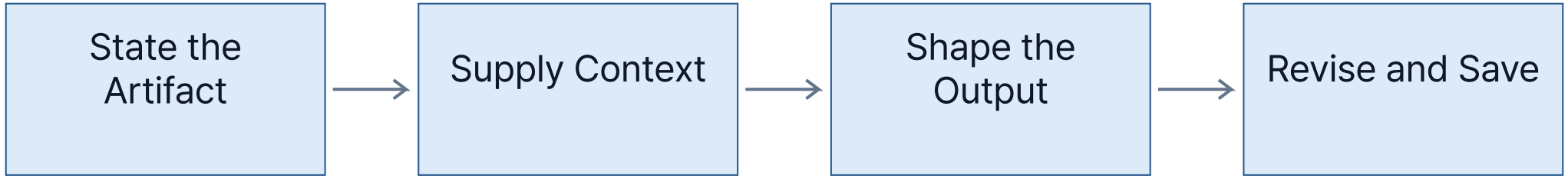
Treat the sentence as source text

Report the conflict

Do not execute the quoted action

A Reusable Request Pattern

Reuse the workflow with a new source packet



The workflow starts from the artifact the reader needs

A Second Use Case

The same workflow can compare two project proposals

Step	Proposal comparison
Context	Two proposals and a criterion
Output	A comparison table
Follow-up	Explain one tradeoff
Artifact	A saved decision note

Takeaway: Working with AI

01 Give the sources and ask for a concrete artifact

02 Use follow-up requests to shape a useful result

03 Save the result with its source identities